

Lucas Alves

MECHANICAL ENGINEERING MASTER'S STUDENT

Vila Nova de Famalicão, Portugal

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Education

Master's in Mechanical Engineering

UNIVERSITY OF PORTO (FEUP)

Porto, Portugal

2026 - Present

Bachelor's in Mechanical Engineering

UNIVERSITY OF PORTO (FEUP)

Porto, Portugal

2023 - 2026

- Final Grade: 17/20 | Final Project Grade: 20/20

Technical Skills

Modeling Tools SolidWorks, Fusion, AutoCAD

CAE & FEM Tools CalculiX, Gmsh, PrePoMax, SimScale

Programming & Automation Python (PyQt, Automation), AI-assisted development

Data & Statistical Analysis MS Excel, JMP, Pandas, Numpy, Matplotlib, PyVista

Experience & Technical Projects

Curricular Internship: Sonotrode Designer App

WONDERMAC INNOVATIVE SOLUTIONS AND UNIVERSITY OF PORTO (FEUP)

February 2026 - July 2026

- Final Project of the Bachelor's in Mechanical Engineering.
- Digitalized and fully automated a CAE pipeline for ultrasonic horn design by developing a Python/PyQt desktop application integrated with Gmsh and CalculiX, applicable to ultrasonic welding and VHCF testing.

Manufacturing Case Study: Drills & Milling

PALBIT

February 2026

- Participated in a technical case study focusing on replaceable-head milling cutting tools.

'Computation Helper' Python Suite

INDEPENDENT GITHUB REPOSITORY

2026 - Present

- Built a suite of digital engineering tools in Python to automate and replace manual computations in practice problems, including spur gear calculations, radiation form factors, and dynamic lookup tables.

Certificates & Coursework

Microeconomics: The Power of Markets

PENN STATE UNIVERSITY

August 2025

- Economic principles and market dynamics, including supply and demand, market structures, and the role of government in markets.

AI for Mechanical Engineering

UNIVERSITY OF MICHIGAN

July 2026 - August 2026

- Consists of three courses in AI for Design and Optimization, Autonomous Vehicles, and Biomedical Applications. Explored Deep Neural Networks and other AI techniques applied to mechanical engineering problems.

Finite Element Analysis Convergence and Mesh Independence

COURSERA

September 2026

- Introduction to convergence and mesh independence in finite element analysis, including practical examples and best practices.

Design of Experiments

UNIVERSITY OF ARIZONA

September 2026 - Present

- Principles and applications of Design of Experiments (DOE) in engineering and research contexts. How to plan, conduct, and analyze experiments to optimize processes and products.

Seminars

Designing High-Performance Medical Device Components for Plastic Welding Assembly

DUKANE CORPORATION

March 2026

Advanced Plastic Welding Technologies for Automotive Applications

DUKANE CORPORATION

April 2026